ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

# ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

## PRACTICE EXAM 5 (PAPER 1) · MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central NC(V) Level 4 Electronics and Digital Concepts for Robotics practice exam 5 (paper 1).**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

**How to mark:** each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable diagrams.

## SECTION A

## QUESTION 1

**1** 1.1 D ✓ 1.2 A ✓ 1.3 C ✓ 1.4 B ✓ 1.5 A ✓ 1.6 D ✓ 1.7 B ✓ 1.8 C ✓ 1.9 A ✓ 1.10 B ✓ (10)

**[10]**

## QUESTION 2

**2** 2.1 True ✓ 2.2 False ✓ 2.3 False ✓ 2.4 True ✓ 2.5 True ✓ 2.6 False ✓ 2.7 True ✓ 2.8 True ✓ 2.9 False ✓ 2.10 True ✓ (10)

**[10]**

## QUESTION 3

**3** 3.1 C ✓ 3.2 I ✓ 3.3 F ✓ 3.4 B ✓ 3.5 H ✓ 3.6 D ✓ 3.7 E ✓ 3.8 J ✓ 3.9 A ✓ 3.10 G ✓ (10)

**[10]**

**TOTAL SECTION A: [30]**

## SECTION B

## QUESTION 4

**4.1.1** $X_L = 2\pi fL = 2\pi \times 50 \times 0{,}2$ ✓ $= 62{,}83\ \Omega$ ✓ (2)

**4.1.2** $Z = \sqrt{(R^2 + {X_L}^2)}$ ✓ $= \sqrt{(40^2 + 62{,}83^2)}$ ✓ $= 74{,}48\ \Omega$ ✓ (3)

**4.1.3** $I = V \div Z = 230 \div 74{,}48$ ✓ $= 3{,}09$ A ✓ (2)

**4.2** Any THREE ✓✓✓: light dimmers; AC motor speed control (fans, drills); AC switching / solid-state relays; heater and temperature control; soft starters. (3)

**[10]**

## QUESTION 5

**5.1** Any TWO losses with how they are reduced, ✓✓ each: copper ($I^2R$) losses in the windings – thicker conductors / low-resistance copper; eddy-current losses in the core – laminated core with insulated laminations; hysteresis losses – core of silicon steel (soft magnetic material). (4)

**5.2** $E_T = 6 \times 1{,}5 = 9\ V$ ✓; $r_T = 6 \times 0{,}1 = 0{,}6\ \Omega$ ✓; $I = 9 \div (5{,}4 + 0{,}6)$ ✓ $= 1{,}5\ A$ ✓; $V = 1{,}5 \times 5{,}4$ ✓ $= 8{,}1\ V$ ✓ (6)

**[10]**

## QUESTION 6

**6.1** An amplifier ✓; positive feedback ✓; a frequency-determining network (RC or LC / crystal) ✓ (3)

**6.2** $f = 1{,}44 \div ((R_1 + 2R_2) \times C)$ ✓ $= 1{,}44 \div ((1\,000 + 20\,000) \times 10 \times 10^{-6})$ ✓✓ $= 6{,}86\ Hz$ ✓ (4)

**6.3** Any THREE ✓✓✓: timers / time delays; switch debouncing; pulse stretching; missing-pulse detectors; touch switches; one-shot triggers for alarms. (3)

**[10]**

## QUESTION 7

**7.1** $R = (V_{in} - V_Z) \div (I_Z + I_L)$ ✓ $= (12 - 5{,}1) \div 0{,}025$ ✓ $= 276\ \Omega$ ✓ (3)

**7.2** When the supply rises, the extra voltage appears across the series resistor ✓; the Zener conducts more current ✓, so the current through the series resistor rises and the voltage drop across it increases, keeping the voltage across the Zener and load at about 5,1 V ✓. (3)

**7.3** Set the meter to the diode test range ✓; red probe on the anode and black on the cathode – a good silicon diode reads about 0,6–0,7 V ✓; reverse the probes – a good diode shows OL (open) ✓; a reading near 0 in both directions means it is shorted, OL in both directions means it is open (faulty) ✓. (4)

**[10]**

## QUESTION 8

**8.1** $V_p = 3{,}5 \times 2$ ✓ $= 7\ V$ ✓ (2)

**8.2** $V_{p\text{-}p} = 2 \times 7$ ✓ $= 14\ V$ ✓ (2)

**8.3** $V_{rms} = 0{,}707 \times 7$ ✓ $= 4{,}95\ V$ ✓ (2)

**8.4** $T = 4 \times 0{,}5$ ✓ $= 2\ ms$ ✓ (2)

**8.5** $f = 1 \div 0{,}002$ ✓ $= 500\ Hz$ ✓ (2)

**[10]**

## QUESTION 9

**9.1** The complement of a sum equals the product of the complements: $\overline{A + B} = \overline{A} \cdot \overline{B}$ ✓✓. The complement of a product equals the sum of the complements: $\overline{A \cdot B} = \overline{A} + \overline{B}$ ✓✓. (4)

**9.2** $F = A(B + \overline{B})$ ✓ $= A \cdot 1 = A$ ✓ (2)

**9.3** Two-input OR gate with inputs A and B ✓; NOT gate on C ✓; two-input AND gate ✓ combining the OR output and $\overline{C}$, output labelled F ✓ (4)

**[10]**

## QUESTION 10

**10.1** A separately excited generator gets its field current from an external DC source ✓✓. A self-excited generator supplies its own field current from its armature (shunt, series or compound connection) ✓✓, starting from residual magnetism. (4)

**10.2** $E_b = V - I_a R_a$ ✓ $= 220 - 20 \times 0{,}5$ ✓ $= 210\ V$ ✓ (3)

**10.3** With no load, the armature current and field flux are very small ✓; the speed of a series motor rises as the flux falls ✓, so without a load it can race to a dangerously high speed and damage itself ✓. (3)

**[10]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**